

```
Downloading: "https://download.pytorch.org/models/alexnet-owt-7be5be79.pth" to /root/.cache/torch/hub/checkpoints/alexnet-owt-7be5be79.pth
100%|██████████| 233M/233M [00:03<00:00, 79.9MB/s]
Downloading: "https://download.pytorch.org/models/vgg16-397923af.pth" to /root/.cache/torch/hub/checkpoints/vgg16-397923af.pth
100%|██████████| 528M/528M [00:07<00:00, 71.5MB/s]
Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to /root/.cache/torch/hub/checkpoints/resnet18-f37072fd.pth
100%|██████████| 44.7M/44.7M [00:00<00:00, 132MB/s]
Downloading: "https://download.pytorch.org/models/googlenet-1378be20.pth" to /root/.cache/torch/hub/checkpoints/googlenet-1378be20.pth
100%|██████████| 49.7M/49.7M [00:00<00:00, 174MB/s]
Downloading: "https://download.pytorch.org/models/mobilenet_v3_large-8738ca79.pth" to /root/.cache/torch/hub/checkpoints/mobilenet_v3_large-8738ca79.pth
100%|██████████| 21.1M/21.1M [00:00<00:00, 73.8MB/s]
Downloading: "https://download.pytorch.org/models/squeezenet1_0-b66bfff10.pth" to /root/.cache/torch/hub/checkpoints/squeezenet1_0-b66bfff10.pth
100%|██████████| 4.78M/4.78M [00:00<00:00, 82.4MB/s]
Downloading http://yann.lecun.com/exdb/mnist/train-images-idx3-ubyte.gz
Failed to download (trying next):
HTTP Error 403: Forbidden
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Downloading https://oss-ci-datasets.s3.amazonaws.com/mnist/train-images-idx3-ubyte.gz
Downloading https://oss-ci-datasets.s3.amazonaws.com/mnist/train-images-idx3-ubyte.gz to ./data/MNIST/raw/train-images-idx3-ubyte.gz
100%|██████████| 9.91M/9.91M [00:00<00:00, 14.5MB/s]
Extracting ./data/MNIST/raw/train-images-idx3-ubyte.gz to ./data/MNIST/raw
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Downloading http://yann.lecun.com/exdb/mnist/train-labels-idx1-ubyte.gz
Failed to download (trying next):
HTTP Error 403: Forbidden
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Downloading https://oss-ci-datasets.s3.amazonaws.com/mnist/train-labels-idx1-ubyte.gz
Downloading https://oss-ci-datasets.s3.amazonaws.com/mnist/train-labels-idx1-ubyte.gz to ./data/MNIST/raw/train-labels-idx1-ubyte.gz
100%|██████████| 28.9k/28.9k [00:00<00:00, 451kB/s]
Extracting ./data/MNIST/raw/train-labels-idx1-ubyte.gz to ./data/MNIST/raw
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Downloading http://yann.lecun.com/exdb/mnist/t10k-images-idx3-ubyte.gz
Failed to download (trying next):
HTTP Error 403: Forbidden
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Downloading https://oss-ci-datasets.s3.amazonaws.com/mnist/t10k-images-idx3-ubyte.gz
Downloading https://oss-ci-datasets.s3.amazonaws.com/mnist/t10k-images-idx3-ubyte.gz to ./data/MNIST/raw/t10k-images-idx3-ubyte.gz
100%|██████████| 1.65M/1.65M [00:00<00:00, 3.98MB/s]
Extracting ./data/MNIST/raw/t10k-images-idx3-ubyte.gz to ./data/MNIST/raw
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Downloading http://yann.lecun.com/exdb/mnist/t10k-labels-idx1-ubyte.gz
Failed to download (trying next):
HTTP Error 403: Forbidden
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Downloading https://oss-ci-datasets.s3.amazonaws.com/mnist/t10k-labels-idx1-ubyte.gz
Downloading https://oss-ci-datasets.s3.amazonaws.com/mnist/t10k-labels-idx1-ubyte.gz to ./data/MNIST/raw/t10k-labels-idx1-ubyte.gz
100%|██████████| 4.54k/4.54k [00:00<00:00, 3.19MB/s]
Extracting ./data/MNIST/raw/t10k-labels-idx1-ubyte.gz to ./data/MNIST/raw
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Downloading http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/train-images-idx3-ubyte.gz
Downloading http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/train-images-idx3-ubyte.gz to ./data/FashionMNIST/raw/train-images-idx3-ubyte.gz
100%|██████████| 26.4M/26.4M [00:02<00:00, 12.4MB/s]
Extracting ./data/FashionMNIST/raw/train-images-idx3-ubyte.gz to ./data/FashionMNIST/raw

Downloading http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/train-labels-idx1-ubyte.gz
Downloading http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/train-labels-idx1-ubyte.gz to ./data/FashionMNIST/raw/train-labels-idx1-ubyte.gz
100%|██████████| 29.5k/29.5k [00:00<00:00, 191kB/s]
Extracting ./data/FashionMNIST/raw/train-labels-idx1-ubyte.gz to ./data/FashionMNIST/raw

Downloading http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/t10k-images-idx3-ubyte.gz
Downloading http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/t10k-images-idx3-ubyte.gz to ./data/FashionMNIST/raw/t10k-images-idx3-ubyte.gz
100%|██████████| 4.42M/4.42M [00:01<00:00, 3.58MB/s]
Extracting ./data/FashionMNIST/raw/t10k-images-idx3-ubyte.gz to ./data/FashionMNIST/raw

Downloading http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/t10k-labels-idx1-ubyte.gz
Downloading http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/t10k-labels-idx1-ubyte.gz to ./data/FashionMNIST/raw/t10k-labels-idx1-ubyte.gz
100%|██████████| 5.15k/5.15k [00:00<00:00, 8.14MB/s]
Extracting ./data/FashionMNIST/raw/t10k-labels-idx1-ubyte.gz to ./data/FashionMNIST/raw

Downloading https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz to ./data/cifar-10-python.tar.gz
100%|██████████| 170M/170M [00:06<00:00, 28.2MB/s]
Extracting ./data/cifar-10-python.tar.gz to ./data
Files already downloaded and verified
Training LeNet-5 on MNIST...
Epoch 1/1, Loss: 0.20785900856654194
Results for LeNet-5 on MNIST:
Accuracy: 0.9726, Precision: 0.9727161715407036, Recall: 0.9726, F1-score: 0.9725457198194388

Training LeNet-5 on FMNIST...
Epoch 1/1, Loss: 0.4459201625065763
Results for LeNet-5 on FMNIST:
Accuracy: 0.873, Precision: 0.872453365864421, Recall: 0.873, F1-score: 0.8724148585795231

Training LeNet-5 on CIFAR-10...
Epoch 1/1, Loss: 1.5220864882402103
Results for LeNet-5 on CIFAR-10:
Accuracy: 0.5468, Precision: 0.5564850350216631, Recall: 0.5468, F1-score: 0.5452174732422953

Training AlexNet on MNIST...
Epoch 1/1, Loss: 0.5830995854608881
Results for AlexNet on MNIST:
Accuracy: 0.9849, Precision: 0.9850122266842704, Recall: 0.9849, F1-score: 0.9848986990054216

Training AlexNet on FMNIST...
Epoch 1/1, Loss: 0.6153614744544029
```

Results for AlexNet on FMNIST:

Accuracy: 0.8483, Precision: 0.861679235077119, Recall: 0.8483, F1-score: 0.8516849857974812

Training AlexNet on CIFAR-10...

Epoch 1/1, Loss: 1.5780851878321078

Results for AlexNet on CIFAR-10:

Accuracy: 0.5374, Precision: 0.5460494832568283, Recall: 0.5374, F1-score: 0.5300588866162591

Training VGGNet on MNIST...

Epoch 1/1, Loss: 0.5369369640136197

Results for VGGNet on MNIST:

Accuracy: 0.9804, Precision: 0.9806243973710908, Recall: 0.9804, F1-score: 0.9804245526306362

Training VGGNet on FMNIST...

Epoch 1/1, Loss: 0.4600168059804419

Results for VGGNet on FMNIST:

Accuracy: 0.8596, Precision: 0.8642248275620436, Recall: 0.8596, F1-score: 0.8587609742974766

Training VGGNet on CIFAR-10...

Epoch 1/1, Loss: 1.524085870484257

Results for VGGNet on CIFAR-10:

Accuracy: 0.5666, Precision: 0.5697899746699551, Recall: 0.5666, F1-score: 0.566066651402662

Training ResNet on MNIST...

Epoch 1/1, Loss: 0.07884356030377808

Results for ResNet on MNIST:

Accuracy: 0.9851, Precision: 0.9852189532694416, Recall: 0.9851, F1-score: 0.985080491865172

Training ResNet on FMNIST...

Epoch 1/1, Loss: 0.3393868381169432

Results for ResNet on FMNIST:

Accuracy: 0.9167, Precision: 0.9196275217598868, Recall: 0.9167, F1-score: 0.9174720690703664

Training ResNet on CIFAR-10...

Epoch 1/1, Loss: 0.8582672437514796

Results for ResNet on CIFAR-10:

Accuracy: 0.7512, Precision: 0.7840016170141964, Recall: 0.7512, F1-score: 0.7488226092559769

Training GoogLeNet on MNIST...

Epoch 1/1, Loss: 0.08274177992923695

Results for GoogLeNet on MNIST:

Accuracy: 0.9892, Precision: 0.9893715957349636, Recall: 0.9892, F1-score: 0.989208943050684

Training GoogLeNet on FMNIST...

Epoch 1/1, Loss: 0.3264627170079807

Results for GoogLeNet on FMNIST:

Accuracy: 0.9129, Precision: 0.9154641198290081, Recall: 0.9129, F1-score: 0.9136014919200081

Training GoogLeNet on CIFAR-10...
Epoch 1/1, Loss: 0.8223028751209264
Results for GoogLeNet on CIFAR-10:
Accuracy: 0.7951, Precision: 0.8038676984930893, Recall: 0.7951, F1-score: 0.7953286517413558

Training Xception on MNIST...
Epoch 1/1, Loss: 0.09122117354107333
Results for Xception on MNIST:
Accuracy: 0.9875, Precision: 0.9877119465430158, Recall: 0.9875, F1-score: 0.9875012028170959

Training Xception on FMNIST...
Epoch 1/1, Loss: 0.3229957341925422
Results for Xception on FMNIST:
Accuracy: 0.9059, Precision: 0.9094119568191551, Recall: 0.9059, F1-score: 0.9027532901868311

Training Xception on CIFAR-10...
Epoch 1/1, Loss: 0.6020004627916514
Results for Xception on CIFAR-10:
Accuracy: 0.8649, Precision: 0.8755245970668766, Recall: 0.8649, F1-score: 0.8674261602763575

Training SENet on MNIST...
Epoch 1/1, Loss: 0.6198520152422506
Results for SENet on MNIST:
Accuracy: 0.9753, Precision: 0.9755299974337808, Recall: 0.9753, F1-score: 0.9753171907729716

Training SENet on FMNIST...
Epoch 1/1, Loss: 0.5585613744313529
Results for SENet on FMNIST:
Accuracy: 0.8527, Precision: 0.8650740898437547, Recall: 0.8527, F1-score: 0.8488089922170197

Training SENet on CIFAR-10...
Epoch 1/1, Loss: 1.5284305759098218
Results for SENet on CIFAR-10:
Accuracy: 0.5345, Precision: 0.5874566093028056, Recall: 0.5345, F1-score: 0.5140610879343797

Model Performance Comparison by Accuracy

